

Suggested problems: 2-7, 13

Problem 1: Imagine a test designed to detect a disease with the following characteristics:

- 1% of the population has the disease.
- If a person has the disease, the test will return a positive 90% of the time.
- If a person does not have the disease, the test will still return a positive 9% of the time.

A patient with no symptoms takes the test and it comes back positive. What is the probability that they actually have the disease?

Problem 2: (Example 2.3.4) Three different machines M_1 , M_2 , and M_3 were used for producing a large batch of similar manufactured items. Suppose that:

- 20% of the items were produced by machine M_1 .
- 30% of the items were produced by machine M_2 .
- 50% of the items were produced by machine M_3 .

Suppose further that:

- 1% of the items produced by machine M_1 are defective.
- 2% of the items produced by machine M_2 are defective.
- 3% of the items produced by machine M_3 are defective.

Finally, suppose that one item is selected at random from the entire batch and it is found to be defective. Determine the probability that this item was produced by machine M_2 .

Problem 3: Suppose that a box contains one fair coin and one coin with a head on each side. Suppose also that one coin is selected at random.

- (a) Suppose that the coin is tossed once and a head is obtained. Determine the probability that the coin is the fair coin.
- (b) Suppose that the coin is tossed a second time and a head is obtained. Determine the new probability that the coin is the fair coin.